EXPERIMENT :18

MATLAB — AI-Based Traffic Prediction for a 5G Network Scenario

OBJECTIVES :

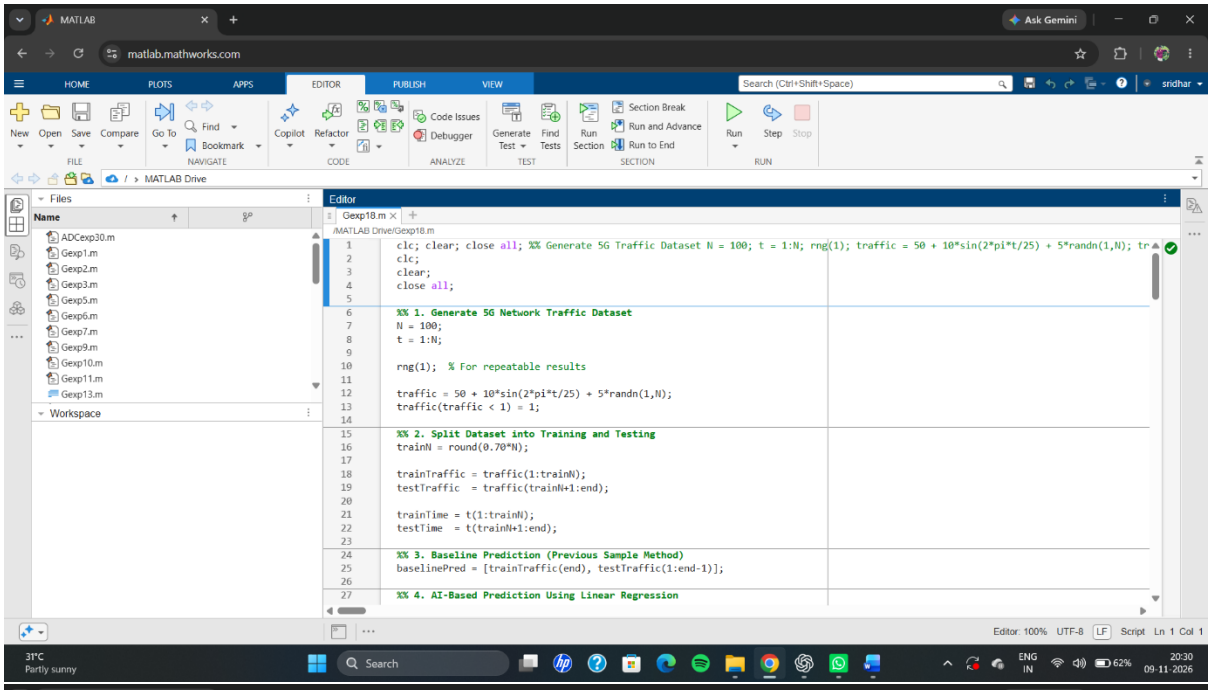
• To load or generate a small time-based 5G network traffic dataset using MATLAB.

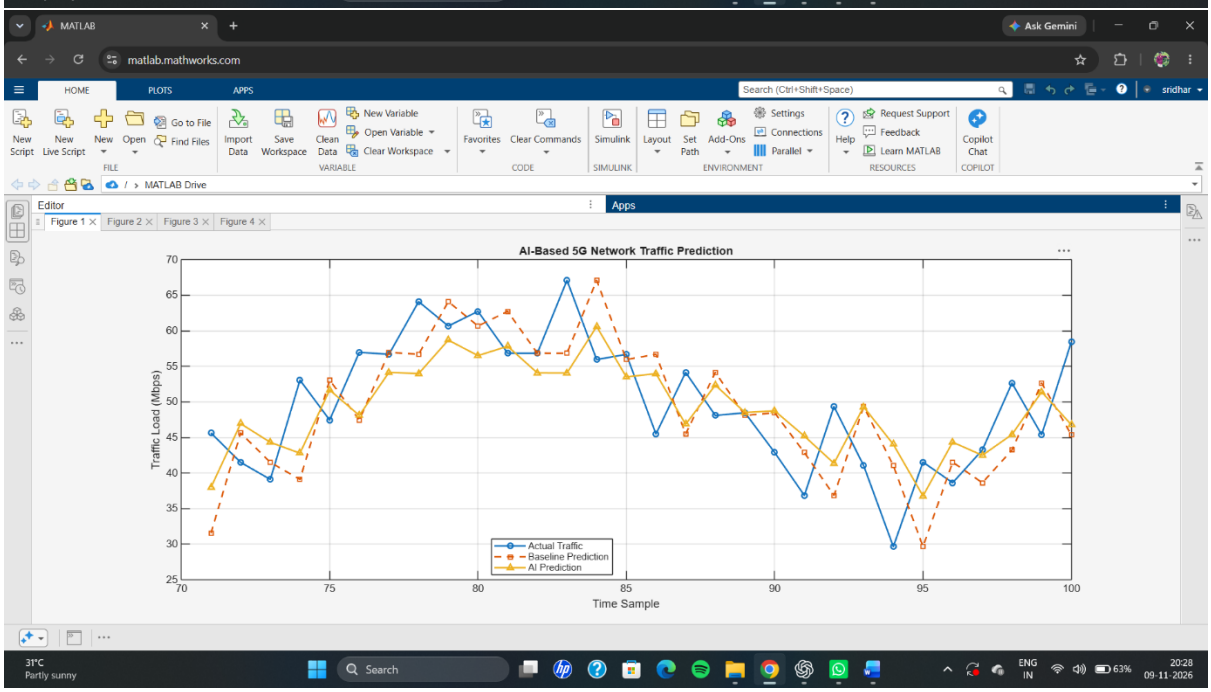
• To develop and execute a basic AI/ML-based traffic prediction model using MATLAB.

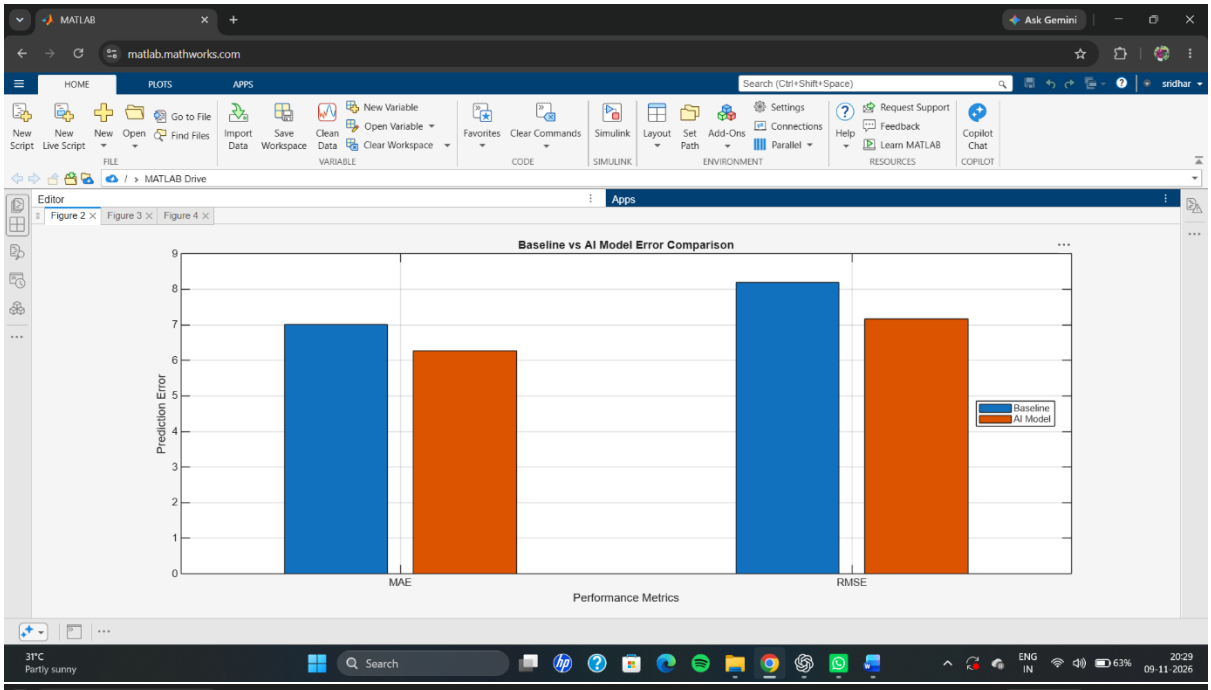
• To compare the prediction performance with a simple baseline method such as previous-sample or mean-value prediction.

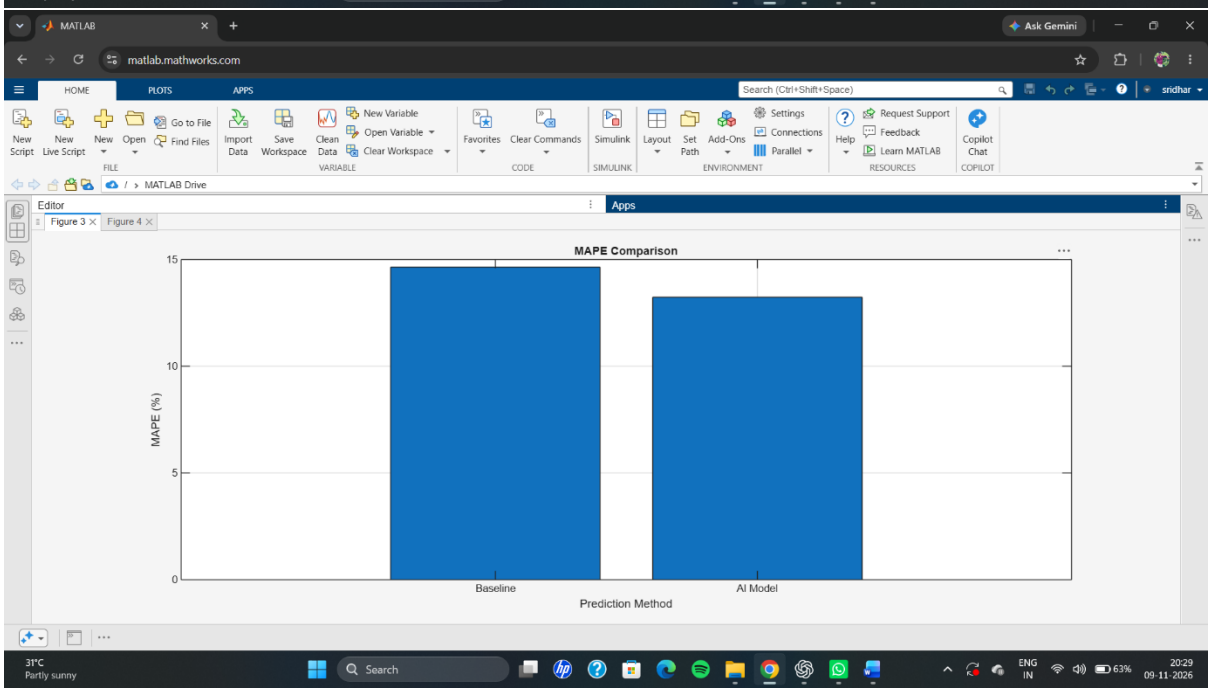
• To analyze how traffic prediction can support radio-resource management, network optimization, and dynamic resource allocation in 5G networks.

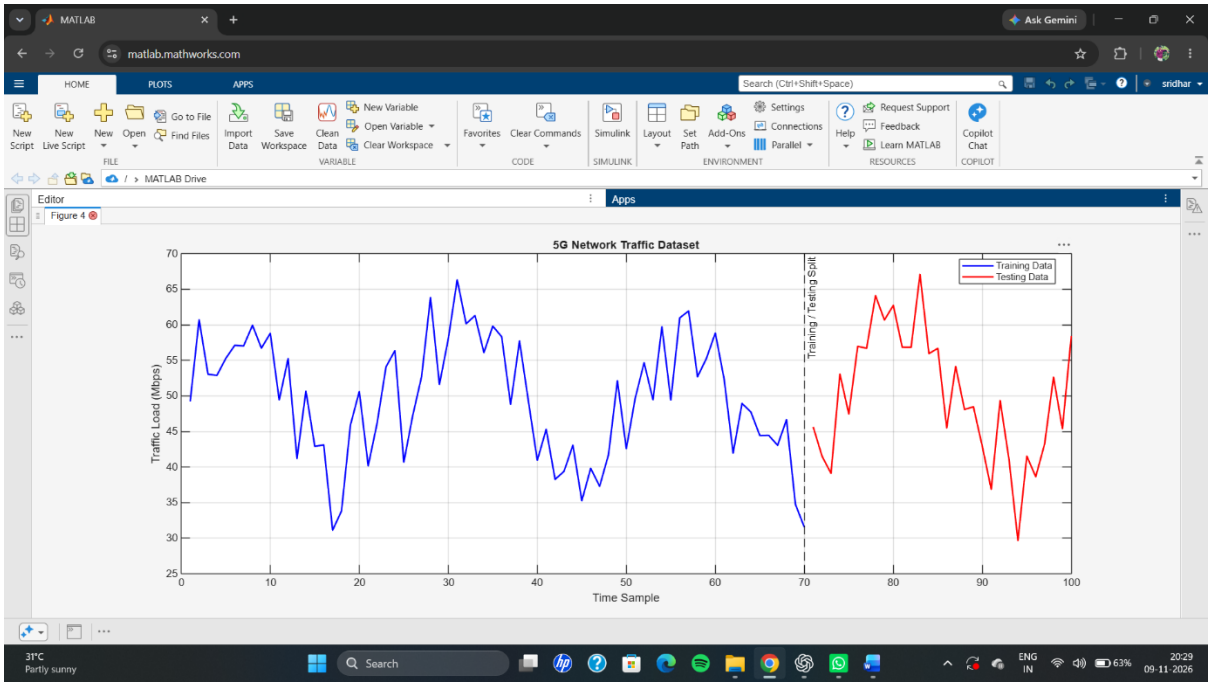
PROGRAME CODE :


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Gexp2.m
Gexp3.m
Gexp5.m
Gexp6.m
Gexp7.m
Gexp9.m
Gexp10.m
Gexp11.m
Gexp13.m
Workspace
Editor
Gexp18.m
/MATLAB Drive/Gexp18.m
clc; clear; close all; %% Generate 5G Traffic Dataset N = 100; t = 1:N; rng(1); traffic = 50 + 10*sin(2*pi*t/25) + 5*randn(1,N); tr
clc;
clear;
close all;
%% 1. Generate 5G Network Traffic Dataset
N = 100;
t = 1:N;
rng(1); % For repeatable results
traffic = 50 + 10*sin(2*pi*t/25) + 5*randn(1,N);
traffic(traffic < 1) = 1;
%% 2. Split Dataset into Training and Testing
trainN = round(0.70*N);
trainTraffic = traffic(1:trainN);
testTraffic = traffic(trainN+1:end);
trainTime = t(1:trainN);
testTime = t(trainN+1:end);
%% 3. Baseline Prediction (Previous Sample Method)
baselinePred = [trainTraffic(end), testTraffic(1:end-1)];
%% 4. AI-Based Prediction Using Linear Regression
Editor: 100% UTF-8 LF Script Ln 1 Col 1
31°C
Partly sunny
Search
ENG
IN
62%
20:30
09-11-2026


MATLAB
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Clean Data
New Variable
Open Variable
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Favorites
Clear Commands
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Add-Ons
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Parallel
Help
Request Support
Feedback
Learn MATLAB
Copilot Chat
FILE
VARIABLE
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SIMULINK
ENVIRONMENT
RESOURCES
COPILOT
MATLAB Drive
Editor
Apps
Figure 1
Figure 2
Figure 3
Figure 4
AI-Based 5G Network Traffic Prediction
Traffic Load (Mbps)
Time Sample
Actual Traffic
Baseline Prediction
AI Prediction
70
65
60
55
50
45
40
35
30
25
75
80
85
90
95
100
31°C
Partly sunny
Search
ENG
IN
63%
20:28
09-11-2026


Baseline vs AI Model Error Comparison
Prediction Error
Performance Metrics
MAE
RMSE
Baseline
AI Model


MAPE Comparison
MAPE (%)
Prediction Method
Baseline
AI Model



RESULT : The MATLAB simulation of AI-Based 5G Network Traffic Prediction was successfully implemented using linear regression. The generated traffic dataset was divided into training and testing data, and traffic predictions were performed using both baseline and AI-based methods. The prediction accuracy was evaluated using MAE, RMSE, and MAPE metrics. The graphical comparison demonstrated the differences between actual and predicted traffic values. This simulation illustrates how AI-based traffic prediction can help improve resource allocation, network performance, and efficient management of 5G communication networks.